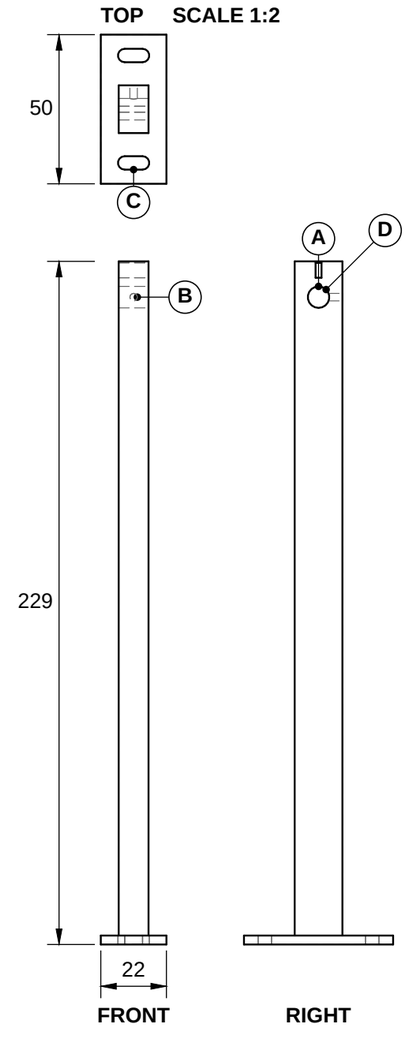
gate_post_L

Printed part (P), qty 1, cad/out/printed/gate_post_L.step. Print: PETG, upright on its foot (as exported), 10 mm brim, 4 walls, 30% infill; needs 230 mm Z height, else print lying on its side with supports

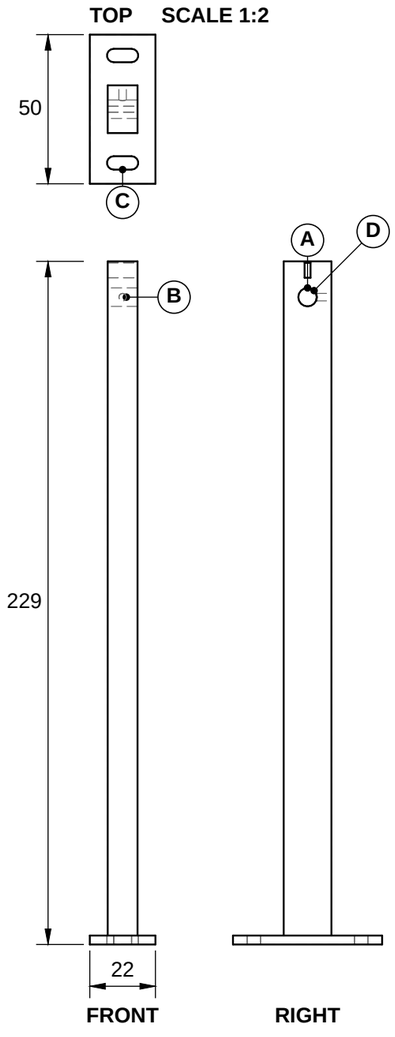
	FEATURES (axes = the STEP file's; positions: see the STEP)
A	Ø7.2 THRU, axis X: laser barrel bore (7 mm Class 2 laser)
B	Ø2.5 from the +Y face into the barrel bore, axis Y: M3 set screw onto the barrel (tap or self-form)
C	2x slot 4.5 wide x 10.5 long, axis Z, centres 36 apart in Y: yaw slot, wood screws
D	Barrel bore axis 217 above the foot underside: on bare plywood the beam is GATE_Z = 214 above the pad top (BUILD_GUIDE 5.9: no shim)



gate_post_R

Printed part (P), qty 1, cad/out/printed/gate_post_R.step. Print: PETG, upright on its foot (as exported), 10 mm brim, 4 walls, 30% infill; needs 230 mm Z height, else print lying on its side with supports

	FEATURES (axes = the STEP file's; positions: see the STEP)
A	Ø6.2 THRU, axis X: receiver barrel bore (6 mm module)
B	Ø2.5 from the +Y face into the barrel bore, axis Y: M3 set screw onto the barrel (tap or self-form)
C	2x slot 4.5 wide x 10.5 long, axis Z, centres 36 apart in Y: yaw slot, wood screws
D	Barrel bore axis 217 above the foot underside: on bare plywood the beam is GATE_Z = 214 above the pad top (BUILD_GUIDE 5.9: no shim)

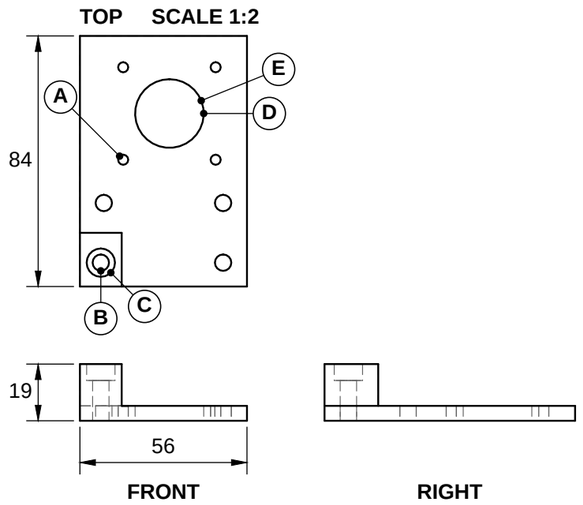


Printed parts are drawn as exported (cad/out/printed/<part>.step): print orientation, bounding-box minimum at the origin; FRONT looks along +Y, TOP down -Z, RIGHT along -X. Dimensions measured on that shape. Drill M3 holes 3.4 and M5 holes 5.5 after printing (BUILD_GUIDE 3.4). Tolerance: see D-00 (PROPOSED).

SlabRail - graded-slab batch scanner	**SHEET D-07 of D-17**
Printed parts: gate posts	SCALE 1:2
Rev A - DRAFT FOR DESIGN REVIEW, not released for fabrication	UNITS: mm 3RD ANGLE
SOURCE: cad/model.py @ 2c95832+uncommitted (drawn by cad/drawings.py)	DATE 2026-10-27 **DO NOT SCALE**

x_motor_mount

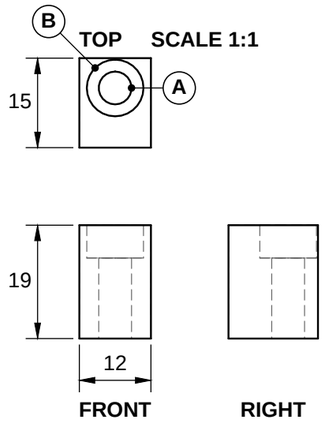
Printed part (P), qty 1, cad/out/printed/x_motor_mount.step. Print: PETG, beam face down (as exported), 5 walls, 40% infill



	FEATURES (axes = the STEP file's; positions: see the STEP)
A	4x Ø3.4 THRU, axis Z, on 31 x 31 (X x Y): M3 clearance
B	4x Ø5.5 THRU, axis Z, centres spread over 41 x 20 (X x Y), not a grid: M5 clearance
C	Ø9.5 x 5.5 deep, axis Z: M5 head, sunk: 1x, at the stop-end M5 hole only
D	Ø23 THRU, axis Z: NEMA 17 pilot clearance
E	Left end = right X hard stop face, 30 from the motor axis; motor axis 20 above the beam top

x_bumper_L

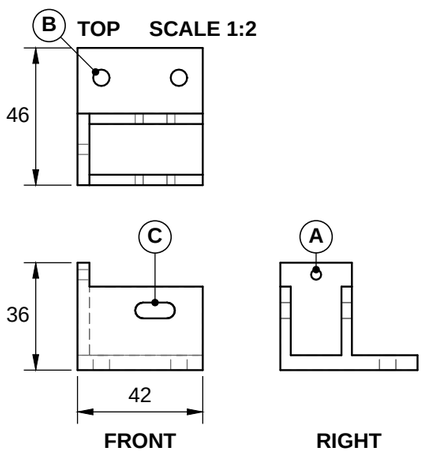
Printed part (P), qty 1, cad/out/printed/x_bumper_L.step. Print: PETG, beam face down (as exported), 4 walls, 40% infill



	FEATURES (axes = the STEP file's; positions: see the STEP)
A	Ø5.5 THRU, axis Z: M5 clearance
B	Ø9.5 x 5.5 deep, axis Z: M5 head, sunk

x_idler_mount

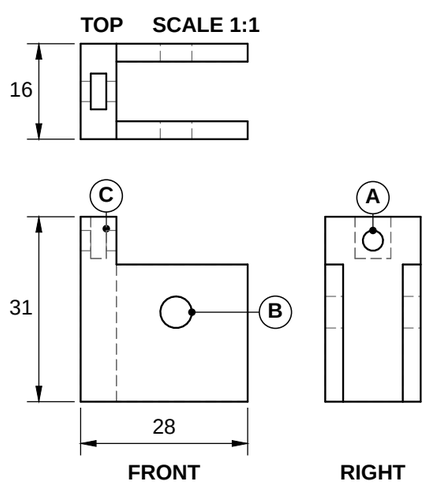
Printed part (P), qty 1, cad/out/printed/x_idler_mount.step. Print: PETG, foot down, 5 walls, 40% infill



	FEATURES (axes = the STEP file's; positions: see the STEP)
A	Ø3.4 THRU, axis X: M3 clearance
B	2x Ø5.5 THRU, axis Z, 26 apart in X: M5 clearance
C	2x slot 5.3 wide x 13.3 long, axis Y: M5 idler axle (tension travel)

x_idler_slider

Printed part (P), qty 1, cad/out/printed/x_idler_slider.step. Print: PETG, as exported, 5 walls, 50% infill



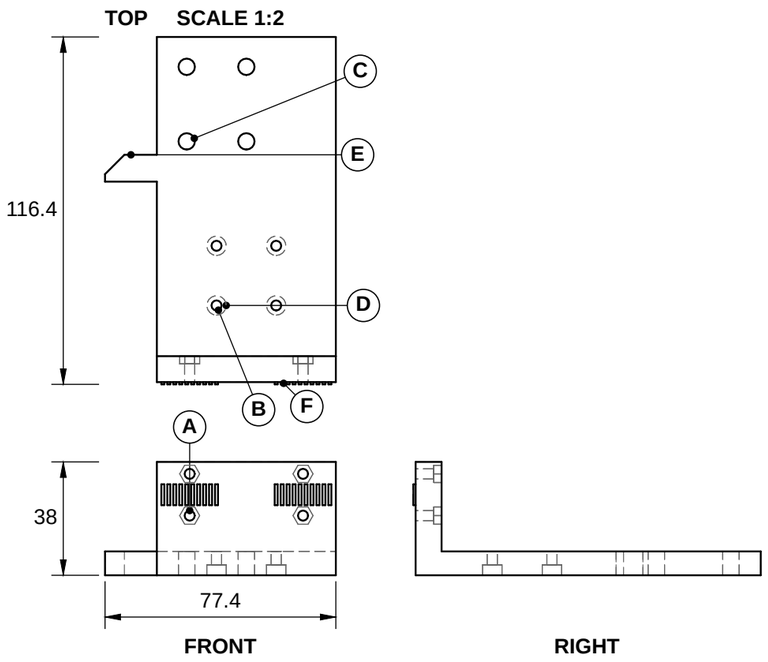
	FEATURES (axes = the STEP file's; positions: see the STEP)
A	Ø3.4 THRU (2 walls on one axis), axis X: M3 clearance
B	Ø5.3 THRU (2 walls on one axis), axis Y: M5 axle
C	M3 nut slot 2.6 wide, from the top, in the bridge

Printed parts are drawn as exported (cad/out/printed/<part>.step): print orientation, bounding-box minimum at the origin; FRONT looks along +Y, TOP down -Z, RIGHT along -X. Dimensions measured on that shape. Drill M3 holes 3.4 and M5 holes 5.5 after printing (BUILD_GUIDE 3.4). Tolerance: see D-00 (PROPOSED).

SlabRail - graded-slab batch scanner	**SHEET D-08 of D-17**
Printed parts: X drive	SCALE 1:1, 1:2 (as noted)
Rev A - DRAFT FOR DESIGN REVIEW, not released for fabrication	UNITS: mm 3RD ANGLE
SOURCE: cad/model.py @ 2c95832+uncommitted (drawn by cad/drawings.py)	DATE 2026-10-27 **DO NOT SCALE**

x_bracket

Printed part (P), qty 1, cad/out/printed/x_bracket.step. Print: PETG, print flat on its spine face (as exported), 6 walls, 50% infill; the finger off its left edge works the X home switch

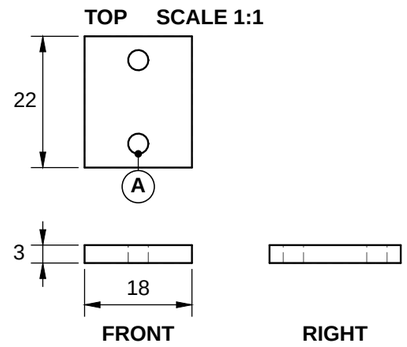


	FEATURES (axes = the STEP file's; positions: see the STEP)
A	4x Ø3.4 THRU, axis Y, on 38 x 14 (X x Z): M3 clearance
B	4x Ø3.4 THRU, axis Z, on 20 x 20 (X x Y): M3 clearance
C	4x Ø5.5 THRU, axis Z, on 20 x 25 (X x Y): M5 clearance
D	4x Ø6.5 x 3.5 deep, axis Z, on 20 x 20 (X x Y): M3 head counterbore
E	X-switch finger 17.4 past the left edge; flat underside at machine Z 369.5; 45° lead-in 6.5 deep
F	GT2 rack on the tab floor, both sides: teeth at 2 pitch (belt teeth down, belt_clamp cap on top)

belt_clamp

Printed part (P), qty 2, cad/out/printed/belt_clamp.step. Print: PETG, flat, 100% infill

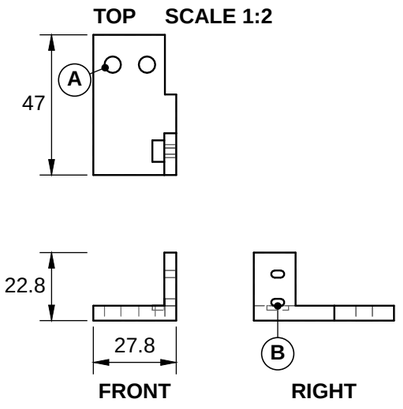
	FEATURES (axes = the STEP file's; positions: see the STEP)
A	2x Ø3.4 THRU, axis Z, 14 apart in Y: M3 clearance



x_switch_mount

Printed part (P), qty 1, cad/out/printed/x_switch_mount.step. Print: PETG, front plate down (as exported), 4 walls, 40% infill; X home switch on 2 x M2x12, 2 x M5x10 button heads

	FEATURES (axes = the STEP file's; positions: see the STEP)
A	2x Ø5.5 THRU, axis Z, 11.5 apart in X: M5 clearance
B	2x slot 2.4 wide x 4.4 long, axis X, centres 9.5 apart in Z: M2

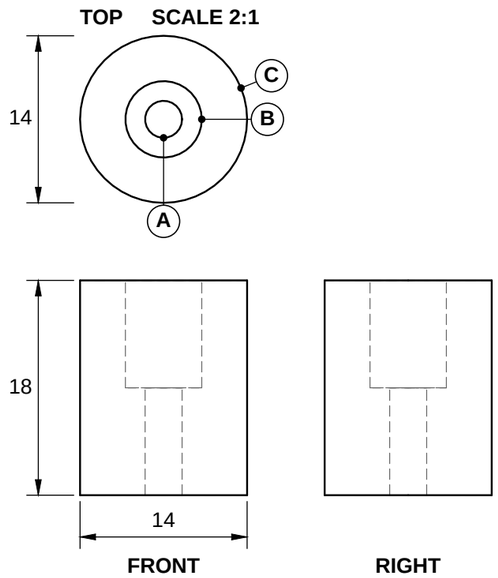


Printed parts are drawn as exported (cad/out/printed/<part>.step): print orientation, bounding-box minimum at the origin; FRONT looks along +Y, TOP down -Z, RIGHT along -X. Dimensions measured on that shape. Drill M3 holes 3.4 and M5 holes 5.5 after printing (BUILD_GUIDE 3.4). Tolerance: see D-00 (PROPOSED).

SlabRail - graded-slab batch scanner	**SHEET D-09 of D-17**
Printed parts: X carriage	SCALE 1:1, 1:2 (as noted)
Rev A - DRAFT FOR DESIGN REVIEW, not released for fabrication	UNITS: mm 3RD ANGLE
SOURCE: cad/model.py @ 2c95832+uncommitted (drawn by cad/drawings.py)	DATE 2026-10-27 **DO NOT SCALE**

awl_adapter

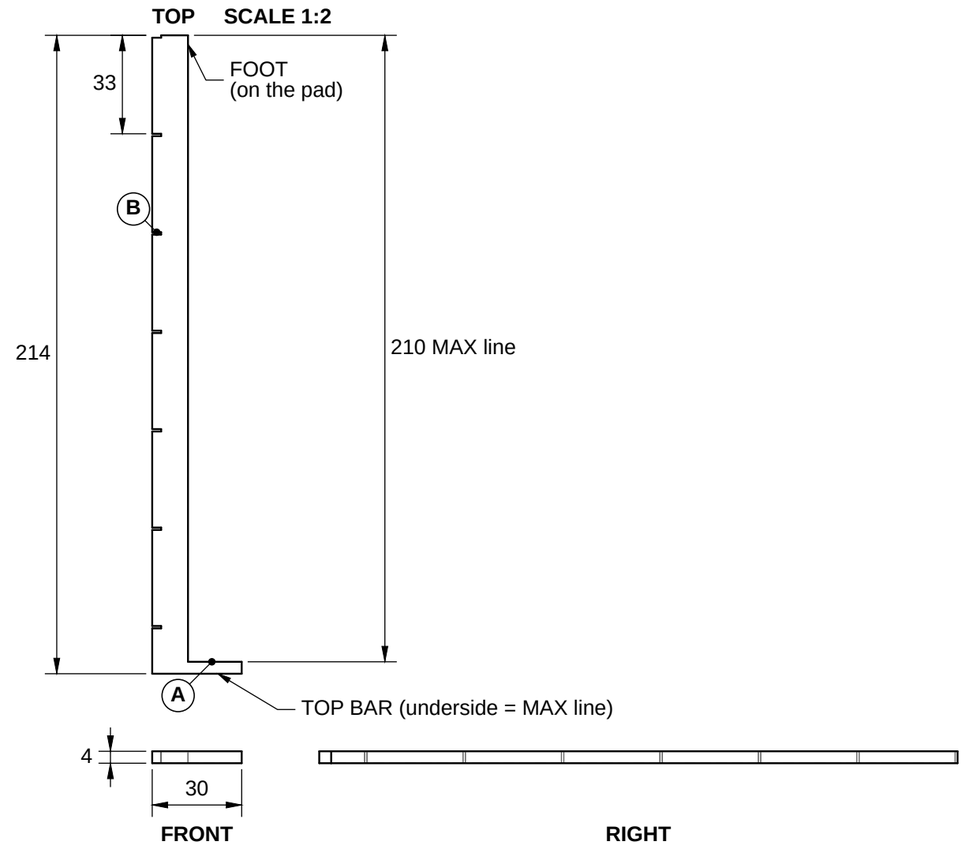
Printed part (P), qty 1, cad/out/printed/awl_adapter.step. Print: PETG, as exported, 100% infill (build tool) + M5 heat-set insert + 3 mm steel pin



	FEATURES (axes = the STEP file's; positions: see the STEP)
A	Ø3.1 THRU, axis Z: 3 mm steel pin, pressed in + CA
B	Ø6.4 x 9 deep, axis Z: M5 heat-set insert pocket
C	Ø14 x 18 long (outside), axis Z: body

fill_gauge

Printed part (P), qty 2, cad/out/printed/fill_gauge.step. Print: PLA, flat (as exported), 3 walls, 20% infill (loading tool)



	FEATURES (axes = the STEP file's; positions: see the STEP)
A	MAX line = underside of the top bar, 210 above the foot (MAX_LINE = 210)
B	Count notches every 33 (5 slabs), 0.8 wide, 3 deep

Printed parts are drawn as exported (cad/out/printed/<part>.step): print orientation, bounding-box minimum at the origin; FRONT looks along +Y, TOP down -Z, RIGHT along -X. Dimensions measured on that shape. Drill M3 holes 3.4 and M5 holes 5.5 after printing (BUILD_GUIDE 3.4). Tolerance: see D-00 (PROPOSED).

SlabRail - graded-slab batch scanner	**SHEET D-14 of D-17**
Printed parts: awl adapter, fill gauge	SCALE 2:1, 1:2 (as noted)
Rev A - DRAFT FOR DESIGN REVIEW, not released for fabrication	UNITS: mm 3RD ANGLE
SOURCE: cad/model.py @ 2c95832+uncommitted (drawn by cad/drawings.py)	DATE 2026-10-27 **DO NOT SCALE**